

ANUP KUMAR JENA

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[LinkedIn](#) [GitHub](#) [Portfolio](#)

SUMMARY

Computer Science undergraduate specializing in Cybersecurity and IoT with hands-on experience building **full-stack web applications using React, Node.js/Express, and FastAPI**. Delivered production-deployed systems with **real-time WebSocket pipelines, REST API design, and Dockerized CI/CD deployments**. Familiar with **microservices architecture, SQL/NoSQL databases, and scalable backend engineering**. Actively learning **Go (Golang)** to expand backend capability. Eager to contribute to **high-throughput, low-latency product development** in a fast-paced startup environment.

TECHNICAL SKILLS

- **Frontend:** React.js, Vite, Tailwind CSS, JavaScript, TypeScript, WebSockets, Three.js, Framer Motion
- **Mobile:** Kotlin, Jetpack Compose, Android (MVVM), Firebase (Auth, Firestore, Storage)
- **Backend & APIs:** Node.js, Express.js, FastAPI, REST API Design, Pydantic, Streamlit
- **Databases:** MySQL, MongoDB / MongoDB Atlas (Motor async driver), SQLite, SQLAlchemy & Alembic, SQL Query Optimization
- **ML & AI:** PyTorch, Scikit-learn, Reinforcement Learning (OpenEnv, policy gradients), LLM API integration (OpenAI, Groq)
- **DevOps & Cloud:** Docker, GitHub Actions, CI/CD Pipelines, Containerization, Vercel
- **Languages:** JavaScript, TypeScript, Python, Kotlin, Java, C, Bash/Shell Scripting
- **Other:** Git, Linux/Unix, IoT Systems (ESP32), structured logging & error handling

PROJECTS

DcoY — Cyber Deception & Active Defense Platform

[GitHub](#) | [Live Demo](#)

Python, FastAPI, React, TypeScript, SQLAlchemy, Scikit-learn, Docker, JWT, WebSockets

- Replaced static threshold alerting with **Isolation Forest** anomaly scoring and automated honeypot misdirection, routing suspicious traffic away from real assets in real time.
- Built a **MITRE ATT&CK**-tagged rule engine and a **Dijkstra-based** attack-path graph that traces compromise propagation across the network topology for faster triage.
- Secured the platform end-to-end with JWT + bcrypt sessions, SHA-256 API keys with constant-time comparison, and request-correlation IDs for full request traceability.
- Shipped with 101 passing Pytest tests, Prometheus metrics, and liveness/readiness probes; deployed via Docker to Cloudflare Pages + Render with Alembic-managed migrations.

TaskMesh — RL-Based Distributed Task Scheduler

[GitHub](#)

Python, PyTorch, FastAPI, OpenEnv, Policy Gradient RL

- Built a custom **OpenEnv** scheduling environment (41-dim observation space, 20-action discrete space) to replace static heuristics (FIFO, SJF) with a learned scheduling policy under bursty, unpredictable workloads.
- Trained a **PyTorch policy-gradient** agent with a shaped reward function (priority-weighted, wait-time-penalized, illegal-action-penalized), reducing average task delay by **15%+** over baseline heuristics.
- Owned backend integration (FastAPI) and core model training as one of 3 engineers; built for OpenEnv Hackathon India 2026.

EcoFlow — Smart Waste Management Platform

[GitHub](#) | [Live Demo](#)

React 19, Node.js, Express, MySQL, WebSockets, Docker, Leaflet, OpenAI API

- Designed a municipal fleet-ops platform with **forced MFA** (TOTP + QR enrollment) and **role-based**

access control with full audit logging, closing gaps left by a single-factor JWT-only baseline.

- Cut route inefficiency by building live route re-optimization on OSRM road geometry, recalculating collection sequences on the fly against adjustable bin-fill thresholds (50–90%).
- Streamed IoT sensor data (fill %, GPS, smoke, tilt) over WebSockets into a filterable Leaflet dashboard, giving operators real-time visibility across the fleet.
- Built a 24-hour predictive waste-accumulation forecast and anomaly-detection service on top of a normalized MySQL schema, with an OpenAI assistant for operator queries.
- Shipped CI/CD via GitHub Actions to Vercel, with Docker for dev/prod parity end to end.

HydroSense — IoT Water Monitoring Platform

[GitHub](#)

Node.js, Express, React, WebSockets, ESP32, Telegram Bot API, Razorpay

- Built an ESP32-to-dashboard telemetry pipeline over WebSockets, giving underground water usage visibility in real time instead of on a delayed billing cycle.
- Enforced tiered daily usage limits (home/apartment/commercial/industry) with timezone-aware resets, and a Razorpay flow for purchasing overage — turning consumption caps into a monetized feature.
- Added Telegram Bot alerts for usage warnings and limit breaches, closing the loop between detection and user notification.
- Led backend development on a 4-person team as part of Samsung Innovation Campus training.

EDUCATION

B.Tech in Computer Science and Engineering (IoT & Cybersecurity)

August 2023 – May 2027

Cambridge Institute of Technology, Bengaluru, India

- Specialization in IoT Systems, Cybersecurity, Linux, Computer Networking, and Distributed Systems.

ACHIEVEMENTS & CERTIFICATIONS

- **OpenEnv India 2026** — Ranked Top 800 among 31,000+ participating teams nationwide.
- **2nd Prize Winner** — INNOVEX 2.0 technical innovation competition.
- **Copado Certified AI Professional** — DevOps and AI automation certification (*Feb 2026*).
- **IBM AI Literacy Certified** — Foundational AI concepts and responsible AI practices.